

dexamine: A Python package for Uniswap event data on Ethereum

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Abstract

Decentralized exchanges record trading and liquidity provision on public blockchains, but empirical analysis requires interpreting these records and linking them to execution metadata. `dexamine` is a Python package that parses Uniswap v2 and v3 events on Ethereum. It converts transaction receipt logs into observations of trades and liquidity changes, with token quantities, pool state, transaction order, and gas information. The package separates data retrieval, contract metadata, protocol interpretation, and output construction. The repository provides recorded Ethereum responses and an offline reproducible example, and version 1 has been used to construct data for two empirical studies of decentralized markets.

Keywords: Ethereum; Uniswap; decentralized finance; market microstructure; research software

1 Motivation and significance

Decentralized exchanges allow users to trade digital assets through programs executed on a blockchain. Uniswap v2 and v3 are automated market makers on Ethereum: trades change the balances of liquidity pools, and prices follow rules implemented by their contracts [1, 2]. Their public records provide data for empirical studies of trading and liquidity provision. However, these records describe contract execution and must be interpreted before they become economic observations.

Studies of decentralized exchange markets examine liquidity and trading costs as well as prices [3, 4]. Research on arbitrage and transaction ordering also shows why execution sequence and fees matter [5]. A transaction’s position within a block, execution fees, and the pool state following a trade help describe the conditions under which it executed. One Ethereum transaction can contain several trades or liquidity events, each of which must remain distinguishable. A block timestamp alone does not recover their execution order.

Ethereum exposes blocks, transactions, and event logs through JSON-RPC. Constructing a dataset requires protocol-specific decoding, resolution of token metadata, conversion of integer quantities into token units, and joins between events and execution metadata. Repeating these transformations across projects creates opportunities for inconsistent units, signs, and ordering. `dexamine` implements them in a reusable Python library. The researcher supplies transaction positions, defined by block number and transaction index, together with a protocol selection and, optionally, a pool address, and receives records for the supported events.

Related tools address extraction and indexing at different levels. Ethereum ETL exports blockchain records and selected token data [6]. cryo supports bulk extraction into tabular formats and event decoding from supplied signatures [7]. Graph Node provides application-specific indexing and GraphQL queries through subgraphs and can be operated locally [8]. Its retained information depends on the chosen schema and mappings. TrueBlocks provides address-based transaction indexing [9]. These tools are discussed as related software; dexamine does not import or invoke them. Researchers can use external tools to select transactions before passing their positions to dexamine.

The contribution of dexamine is a common representation of Uniswap events joined to their execution metadata. Generic extraction tools still require these transformations, while a subgraph places them within an indexing service. A separate library allows researchers to reuse RPC and contract-access facilities while incorporating the protocol interpretation into their own pipelines. Transaction discovery, storage, and statistical estimation remain independent stages. Version 1 of dexamine has been used for data construction in two studies of decentralized markets [10, 11].

The software, documentation, tests, and reproducible examples are available at <https://github.com/HanssonMagnus/dexamine>.

2 Software design and implementation

2.1 Installation and usage

dexamine can be installed directly from the source repository:

```
python -m pip install git+https://github.com/HanssonMagnus/dexamine.git@v1.1.0
```

The package requires Python 3.10 or newer, web3.py 6.15.0 or newer, and Requests 2.31.0 or newer. Live parsing requires access to an Ethereum JSON-RPC endpoint retaining the requested historical blocks and transaction receipts and supporting contract metadata calls. The recorded example in Section 3 runs offline without a node.

2.2 Software architecture

Figure 1 shows the separation of JSON-RPC retrieval, contract metadata resolution, protocol parsers, and output construction. A reusable session loads contract interfaces and retains pool and token metadata across calls. dexamine’s JSON-RPC client uses Requests [12] to retrieve transactions, receipts, and blocks over HTTP. web3.py [13] supplies contract calls for pool and token metadata and address checksum conversion. Requests and web3.py are the two declared runtime dependencies. The protocol parsers decode event topics and fixed-width data fields directly in Python. The output layer joins these event records to execution metadata.

Requests for multiple positions are batched. Results are yielded incrementally, so event records need not accumulate in memory. The metadata cache grows with the number of distinct contracts encountered. Users can divide positions across processes with one session per worker. Processing rates depend on endpoint latency and capacity, batch size, and metadata cache misses; no throughput advantage over other extraction tools is claimed here.

2.3 Software functionalities

The package supports Uniswap v2 and v3 swaps, mints, and burns on Ethereum mainnet. Mints and burns describe additions and removals of liquidity. Event records include token symbols and decimal precision, normalized token quantities, and event-specific pool information. The sign

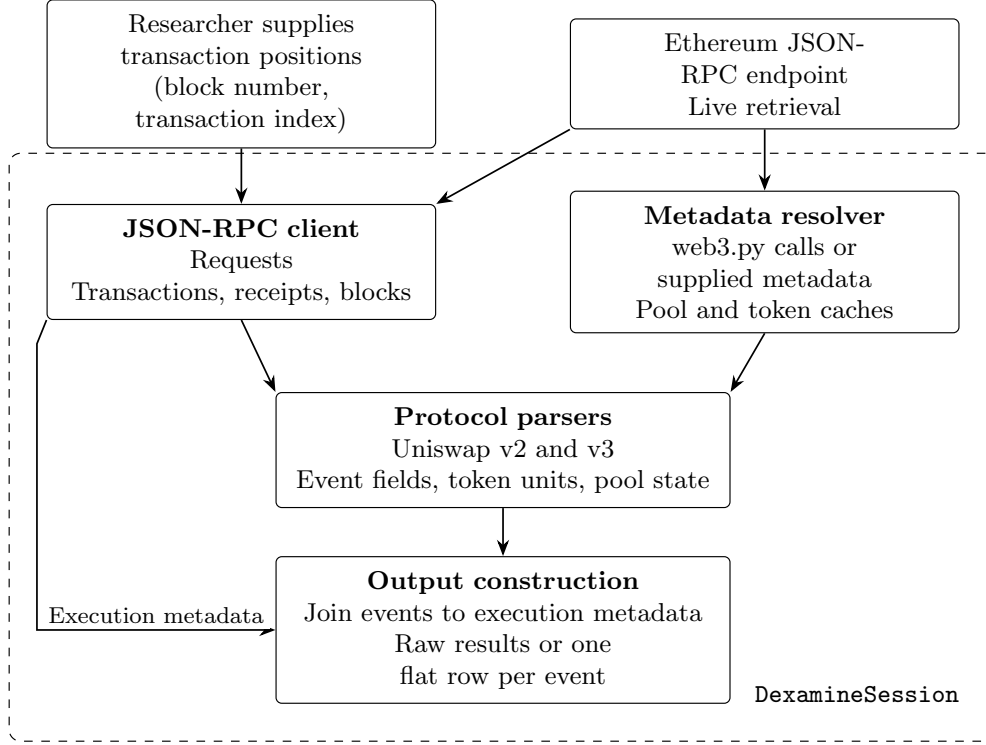


Figure 1: Data flow within a session. The RPC client retrieves execution records; the metadata resolver supplies pool and token information. Protocol parsers decode receipt logs, and the output layer joins the resulting events to transaction and block metadata. Offline replay supplies recorded RPC results and seeded metadata.

conventions express liquidity additions as positive quantities and removals as negative quantities; swap quantities describe the pool’s net token flows.

The parsers account for protocol differences. Uniswap v2 reports reserve updates in a preceding **Sync** event. `dexamine` checks that the immediately preceding log is a **Sync** from the same pool and skips an event if this condition fails. For swaps, the reserve ratio supplies the pool’s post-event marginal price before fees. Uniswap v3 swap logs report price, tick, and active liquidity directly. The parser uses these values to derive price and virtual reserves. These reserves describe the local trading curve, rather than total pool balances; they are reported as missing when active liquidity is zero.

The output can retain node payloads alongside parsed events or provide one flat row per event. Flat records include block and transaction indices, the transaction hash, and the event’s zero-based position within the receipt’s log list. This last field is named `receipt_log_index`; it differs from Ethereum’s block-wide `logIndex`. Together, transaction identity and receipt position identify the source event. Missing or inapplicable numerical fields are represented by `None`, which becomes `null` in JSON.

Token amounts are scaled by their decimal precision. Normalized quantities use floating-point arithmetic and can contain rounding error, so the output is intended for empirical analysis rather than exact integer accounting. Retaining the original logs allows users to recover integer quantities. Gas consumption and fee fields apply to the entire transaction and are repeated across its events. They do not allocate execution costs to individual trades.

Destination labels are constructed from the transaction’s top-level destination. Transactions with no destination are labeled `contract_creation`; destinations matching a bundled list of

Uniswap router addresses are labeled `uniswap_router`; and all remaining destinations are labeled `other_contract`. This classification does not reconstruct internal call paths or establish trader identity, arbitrage, or maximal extractable value.

2.4 Reproducibility and verification

Reproducibility depends on the software version, source responses, and resolved metadata. Pool and token metadata are queried at the latest block and cached. Changes to token metadata can therefore affect repeated runs even when historical logs are unchanged. Retaining input responses and metadata supports replication. The public `MetadataResolver.seed` method accepts recorded token and pool metadata, normalizes addresses, and replaces the supplied cache entries. Missing entries still trigger endpoint queries; callers retain responsibility for the provenance of supplied metadata. The endpoint must serve the requested historical blocks and receipts and support metadata calls, but the parser does not require historical contract-state queries. Users should also retain transaction hashes when selecting positions, since chain reorganizations can change the transaction at a recent position.

Offline tests exercise event decoding, signed quantities, reserve ordering, pool filtering, and missing values using recorded responses and constructed event logs. An optional integration suite checks the public interface against a live endpoint. Continuous integration runs tests, formatting, linting, and strict type checking on Python 3.10 through 3.13. Installation instructions, field definitions, limitations, and contribution guidance are supplied with the code.

3 Illustrative examples

Transaction 31 in Ethereum block 12,561,528, recorded on 3 June 2021, contains four Uniswap v3 swaps across three pools. The following code parses all supported v3 events in the transaction without restricting the pool address, then orders the rows by their positions in the receipt:

```
from dexamine import DexamineSession

session = DexamineSession.from_node_url(node_url)
rows = session.parse_position(
    block_number=12561528,
    tx_index=31,
    protocol="uniswap_v3",
    exchange_pair_address=None,
    output_format="flat",
)
rows.sort(key=lambda row: row["receipt_log_index"])
```

Table 1 shows the four swaps in receipt order. The token symbols follow each pool's token0/token1 ordering. Positive quantities enter the pool; negative quantities leave it. USDC amounts are scaled by 10^6 , and DAI and WETH amounts by 10^{18} . The intervening logs include token transfers and other events outside the v3 parser's supported set. Their presence explains the gaps between receipt positions. The first and last swaps concern the same USDC/WETH pool, but remain distinct observations within one transaction.

Each swap also reports the post-event price, tick, and active liquidity. Table 2 shows the price and virtual reserves derived from that state. Prices are expressed as token 0 per token 1: USDC per WETH, DAI per USDC, and DAI per WETH for the respective pools. These are marginal prices after the swap, rather than ratios of the exchanged amounts. Virtual reserves characterize

Receipt position	Token 0	Token 1	Amount 0	Amount 1
3	USDC	WETH	-14008.253925	5.000000
6	DAI	USDC	-13983.724002	14008.253925
10	DAI	WETH	13983.724002	-4.999395
13	USDC	WETH	-14003.702900	4.999395

Table 1: All four Uniswap v3 swaps in the recorded transaction. Amounts are in token units and rounded to six decimal places. Receipt positions are zero-based.

the local trading curve at the reported active liquidity; they are not the pool’s total token balances. In particular, the large DAI/USDC virtual reserves should not be interpreted as assets held by that pool.

Receipt position	Price	Tick	Virtual reserve 0	Virtual reserve 1
3	2802.766431	196936	137336604.311	49000.374
6	0.998748	-276312	94086697387.618	94204630210.412
10	2800.001801	-79378	9064867.550	3237.451
13	2802.194884	196938	137322600.609	49005.371

Table 2: Post-swap pool state for the events in Table 1. Prices are rounded to six decimal places and virtual reserves to three, in the corresponding token units. Active liquidity and the encoded square-root price are retained in the complete JSON output.

The explicit sort recovers execution order even when different supported event types are grouped by the parser. Pool and token addresses are retained in the recorded inputs; symbols provide readable labels, not unique contract identifiers. Gas usage of 476,588 and an effective gas price of 40 gwei apply to the whole transaction and repeat on every row. Summing those fields across swaps would count the same transaction cost four times. The `other_contract` destination label describes the top-level address match and does not establish trader identity or the transaction’s economic purpose.

The accompanying recorded example reproduces the output offline with dexamine 1.1.0. It contains the full transaction, receipt, and block responses, the transaction hash and capture provenance, and metadata for all three pools and tokens. The metadata was captured separately through contract calls at the recorded block, rather than at the latest block. The script supplies it through `MetadataResolver.seed` and replays the public RPC client’s `call` method while exercising the session, parser, and output construction. Unexpected HTTP access is rejected.

The `--check` option compares all four rows with the committed JSON; `--node-url` enables live retrieval with normal metadata resolution. A separate verification script checks transaction and block identity, decodes the archived metadata calls, and independently checks each swap’s amounts and pool state using ABI decoding and high-precision decimal arithmetic. Both checks run in continuous integration without a node.

4 Impact

Version 1 of dexamine was used to construct data for *Price Discovery in Constant Product Markets* [10] and *Measuring DeFi Price Impact and A New Empirical Market Microstructure Model* [11]. The former studies the contribution of trading and liquidity provision to price discovery on Uniswap. The latter develops an empirical market microstructure model for decentralized exchanges that distinguishes arbitrageurs when analyzing returns and order flow. These applications use the parser as a data-construction component; trader classification and

econometric estimation are performed separately.

Both applications require preserving the sequence of events and their associated pool information. Reusing a parser provides consistent treatment of token units, liquidity changes, and execution metadata across analyses. It also makes the data transformations inspectable independently of the statistical model. The research use of version 1 predates the first public release. This paper describes version 1.1.0, which adds the public metadata-seeding interface used by the offline example.

The same representation supports further studies of liquidity provision, execution costs, and price adjustment. Researchers can select different transaction samples or apply alternative classifications while retaining the event construction. For example, a study of gas costs can aggregate at transaction level before linking costs to a set of events, while a study of liquidity can retain separate mint and burn observations.

5 Conclusions

dexamine converts Uniswap v2 and v3 receipt logs into event observations linked to Ethereum execution metadata. Its contribution is a reusable implementation of protocol interpretation and data joins for empirical research. The library supports both individual transactions and batched processing, with offline verification and an executable example. Its scope and numerical conventions are explicit: transaction discovery, trader attribution, and statistical analysis remain the responsibility of the surrounding research workflow.

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Code and data availability

The source code for dexamine is available at <https://github.com/HanssonMagnus/dexamine> under the GNU General Public License v3.0 or later. This paper describes version v1.1.0.

The recorded Ethereum responses used in Section 3 are included under `paper/examples/data`. The example scripts, expected output, historical metadata, and recorded inputs are distributed with the repository. The permanent example snapshot contains the files needed to reproduce all four swaps with dexamine 1.1.0. They are also supplied as ancillary material with this preprint.

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